Bias Variance Trade-Off.

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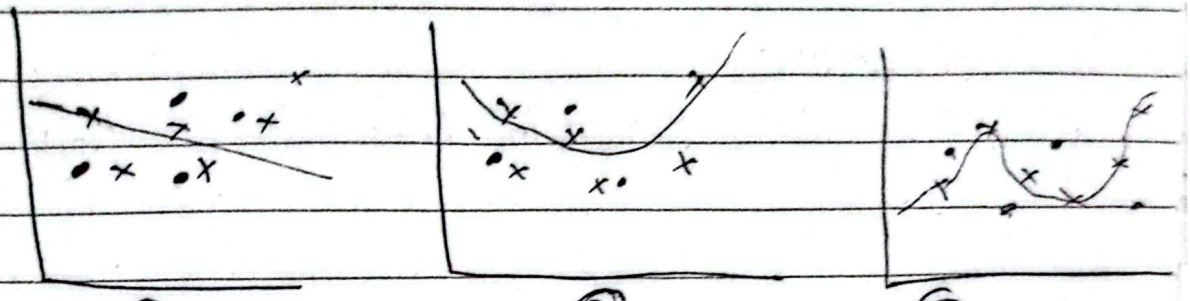
Bias = the inability of the ml model to truely capture the relationship in train data.



① High Bias Low Variance

② Low Bias Low Variance.

③ Low Bias High Variance

- Fig: ① has high biasness.

Variance : diff of fits in diff datasets.

Case I:

train data error: 90%
test data error : 10%
variance = |90% - 10%|
= 80%
High ↑

Case II:

test data error: 15%
train data error = 10%
variance = |15% - 10%|
= 5%
Low ↓

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Overfitting:

model works better in train data ↔ low error in train
but can't work better in test data ↔ high error in test
Fig ③.

Underfitting:

- Can't work better in train data ↔ high error in train
Fig ①.

~~Types of Regularizatio~~

~~RIDGE REGUARIZATION~~

Regularization:

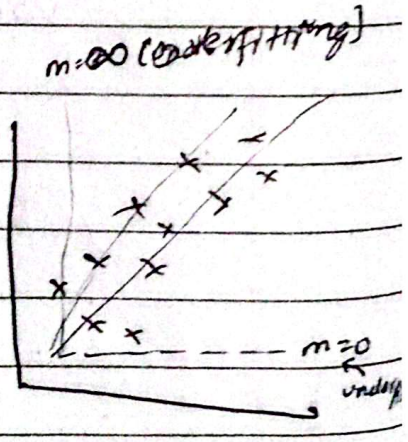
- add some ~~adape~~ term in the loss function to decrease over fitting.

Types: 1) Ridge (L2)
2) LASSO (L1)
3) Elastic Net. [Ridge + LASSO].

1) Ridge (L2)

~~In ove~~ Let $y = mx + b$

- If value of m is extermely high, then overfitting
- Under fitting → low m.



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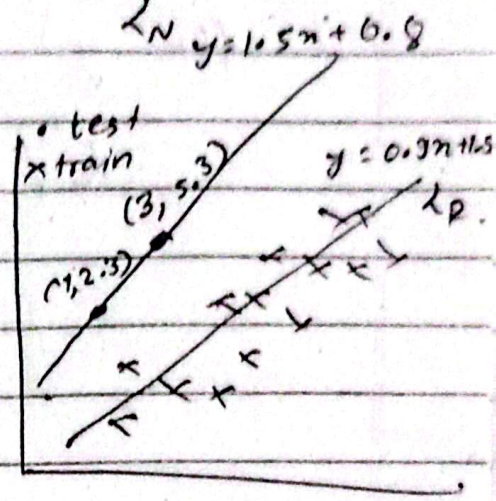
- we have to do little bit low value of m for regularization.



Here we have to choose L_R.

$$L = \frac{\sum_{i=1}^{n} (y_i - \hat{y}_i)^2}{n} + L(m^2)$$

Here, L = hyper parameter. $[L \in (0, \infty)]$.

step 1: Calculate loss for both lines.

Let $L = 1$ [for now].

Loss L_N	Loss L_R
$0 + 1.5^2$	$(2.9 - 0.9 - 1.5)^2 +$
$= 2.25$.	$(5.2 - 2.7 - 1.5)^2$.
	$= 0.1^2 + 1.1^2 + 0.9^2$
	$= 2.03$.

so, ml will choose L_R coz of regularization.